

01204211 Discrete Mathematics

Lecture 12a: Undecidability (1)

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September 15, 2026

Non-context-free languages

While

$$\{0^n 1^n \mid n \geq 0\}$$

is context free, the language

$$\{0^n 1^n 0^n \mid n \geq 0\}$$

is not.

Non-context-free languages

While

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is context free, the language

$$\{0^n 1^n 0^n \mid n \geq 0\}$$

is not.

Can we write a python program to check if a string w belongs to the language $\{0^n 1^n 0^n \mid n \geq 0\}$?

Grammars

reg ex
regular language

Context-free
grammars

CFL

$\{0^n 1^n 0^n \mid n \geq 0\}$

DFA

↓ nondeterminism
NFA

memory: stack

NFA + stack

stack

memory: random-access

• Undecidability
↑ concept

reduction

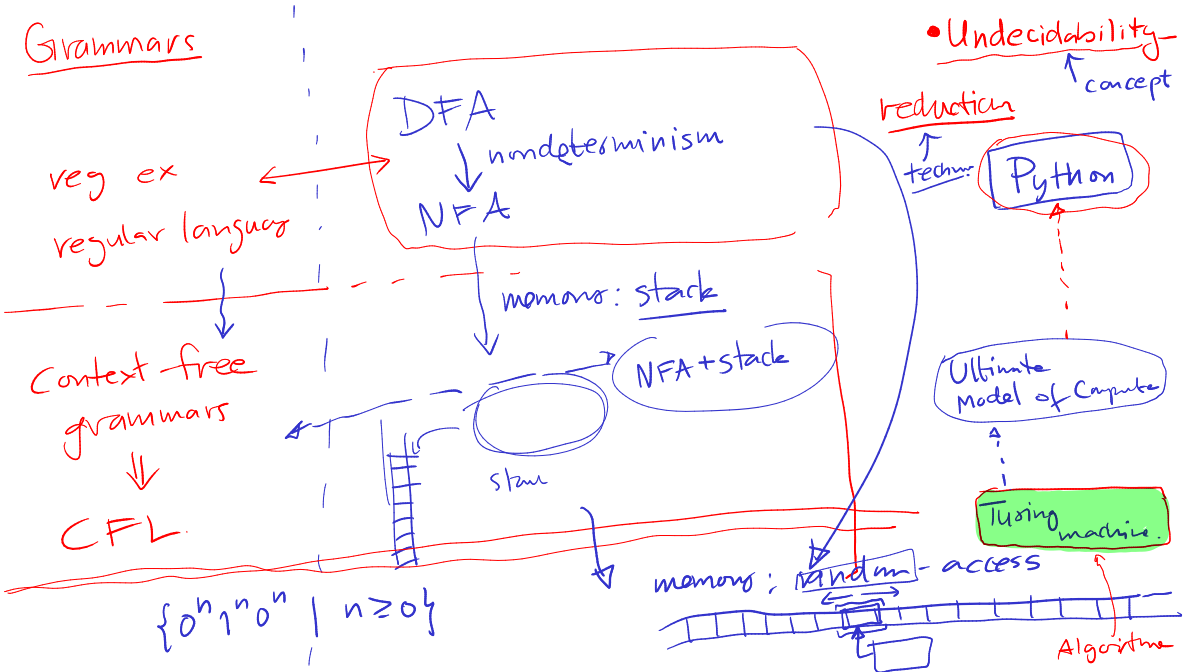
↑ technique

Python

Ultimate
Model of Compute

Turing machine

Algorithm



$$\{0^n 1^n 0^n \mid n \geq 0\}$$

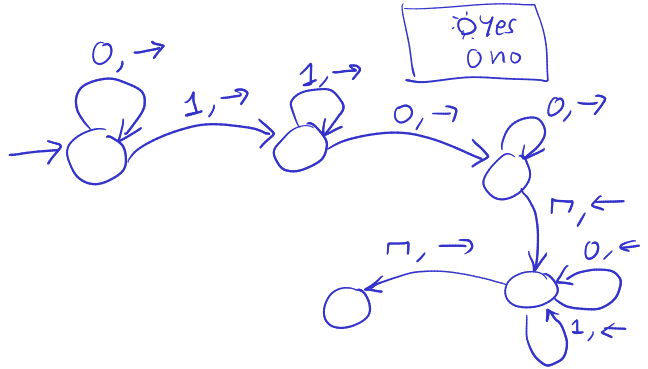


Q. answer yes

1. check if string is $0^n 1^n 0^n$

1.1

$$0^* 1^* 0^*$$



2.

Big question

Is there a python program that “solves” any possible problem?

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Can a computer solve any problem?

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Can a computer solve any problem?

Is there an algorithm that solves every problem?

Big question

Is there a python program that “solves” any possible problem?

Can a computer solve any problem?

Is there an algorithm that solves every problem?

What is the limit of computation?

Answer by a counting argument

If there are “more” problems than any possible algorithms, then there should be some problem that algorithms cannot solve.

(Think of an algorithm as “a python program.”)

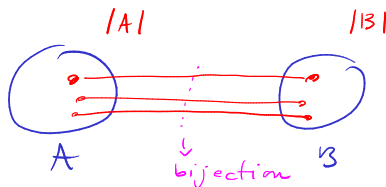
Answer by a counting argument

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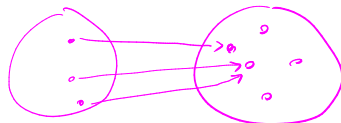
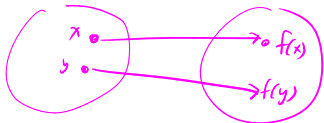
However, there are infinitely many python programs and there are infinitely many problems. It is not obvious how to make such an argument formally.

Bijections



Definition

- ▶ A function $f: A \rightarrow B$ from domain A to range B is **one-to-one** if for any $x \neq y \in A$, $f(x) \neq f(y)$.
- ▶ A function $f: A \rightarrow B$ from domain A to range B is **onto** if for any $x' \in B$, there exists $x \in A$ such that $f(x) = x'$.
- ▶ A function $f: A \rightarrow B$ is a **bijection** (or bijective) if it is one-to-one and onto.



Bijection: examples

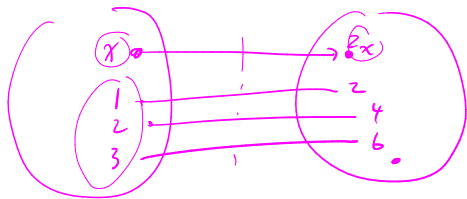
$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

$$\mathbb{E} \subseteq \mathbb{N}$$

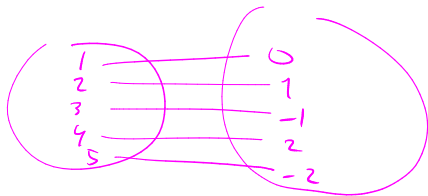
$$\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$$

$$\mathbb{E} = \{2, 4, 6, 8, \dots\}$$

$$\mathbb{E} \subset \mathbb{N}$$



$f(x) = 2x$ is a bijection



Lemma 1

✕ For any set A , there is no bijective function $f : A \rightarrow 2^A$.

Proof.

We prove by contradiction. Assume that there exists a bijective function f from A to 2^A . We construct a set $B \subseteq A$ such that there is no $x \in A$ such that $f(x) = B$.

$$B \in 2^A$$

Lemma 1

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$$B = \{x \in A \mid x \notin f(x)\}.$$

Now suppose that there exists $x \in A$ such that $f(x) = B$. There are two cases to consider:

Case 1: $x \in B$. $\Rightarrow x \in f(x) \Rightarrow x \notin B \quad \times$

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Case 1: $x \in B$.

Case 2: $x \notin B$. $x \notin f(x) \Rightarrow x \in B \quad \times$

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Now suppose that there exists $x \in A$ such that $f(x) = B$. There are two cases to consider:

Case 1: $x \in B$. ✗

Case 2: $x \notin B$. ✗

In both case, we have a contradiction; therefore, our assumption is false. Thus, there is no bijection between A and 2^A . □

Example: finite set

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ Consider function $f : A \rightarrow 2^A$ defined as

$$f(1) = \{\}$$

$$f(2) = \{1, 2, 3\}$$

$$f(3) = \{1, 2, 3, 4, 5, 6, 7\}$$

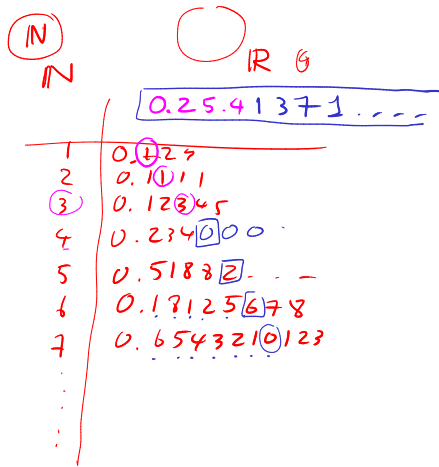
$$f(4) = \{1, 3, 5, 7\}$$

$$f(5) = \{2, 4, 6\}$$

$$f(6) = \{7\}$$

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$B =$



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	1	2	3	4	5	6	7
1							
2	/		/				
3	/	/	/	/	/	/	/
4	/		/	/	/		/
5		/		/	/	/	
6						/	/
7	/	/	/				/

↑ Diagonalist

$B =$

$$f(3) \stackrel{?}{=} B$$

$$\{1, \square, \square, 4, 5, 6, 7\}$$

Example: infinite set

Let $A = \mathbb{N} = \{1, 2, 3, 4, \dots\}$. Consider function $f : A \longrightarrow 2^A$ defined as

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$$f(5) = \{2, 4, 6, \dots\}$$

$$f(6) = \{7\}$$

$$f(7) = \{1, 2, 3, 11, 12, 13, 21, 22, 23, \dots\}$$

$$\vdots$$

$$B =$$

	1	2	3	4	5	6	7	...
1								
2								
3								
4								
5								
6								
7								
$\vdots$								

$$\text{bijection } A \sim 2^A$$

The previous lemma informally states that there are “more” subsets than the number of elements in the set.

Let's think about:

- ▶ A set of all python programs, and
- ▶ A set of all languages.

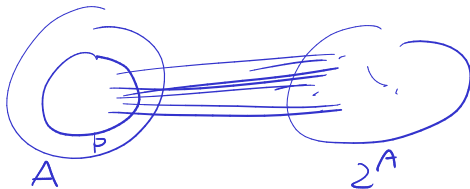
$$= 2^A$$

$$P \subseteq A$$

= form an expression $\{0,1\}^*$
= sub

$$2^A$$

= form an subset of A
= set of all languages



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Since each python program “solves” at most one language, there are not “enough” python programs to solve all possible language.

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- ▶ A set of all languages.

Since each python program “solves” at most one language, there are not “enough” python programs to solve all possible language.

But what exactly is a problem that cannot be “solved”?

- * • ปัญหา python สามารถหาคำตอบได้หรือไม่?
- ปัญหา “หาคำตอบ” ที่ ไม่สามารถหาด้วย python ที่ “solve” ได้. ?

Decision problems

- ▶ Given an integer x , is x odd?
- ▶ Given a string w , is w palindrome?
- ▶ Given a string w , is $w \in \{0^n 1^n \mid n \geq 0\}$?
- ▶ Given a map, a starting position s , a destination t , and an integer k , does there exist a path from s to t with distance at most k ?
- ▶ Given a program P and input string w , when running P with w as an input, does P terminate?

$\{0^n 1^n \mid n \geq 0\}$



$\{ \langle P, w \rangle \mid P \text{ does not terminate on } w \}$

Decision problems and languages

For this problem:

Given an integer x , is x odd?

we can define a corresponding language

$$L_E = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\}.$$

To solve this problem, given x , we can ask if $x \in L_E$.

Languages and programs

We will talk about languages of particular programs. For example, let $\mathbb{P}$ be the set of all python programs. In this case, $\mathbb{P}$ is a language.

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$$\{(P, Q, x) \mid P, Q \in \mathbb{P}, P(x) \text{ and } Q(x) \text{ terminate with the same output.}\}$$

Programs and inputs

```
x = int(input())
if x % 2 == 0:
    print('yes')
else:
    print('no')
```


Programs and inputs

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```

```
$ python le.py
10
yes
$ python le.py
7
no
```

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Traceback (most recent call last):
  File "le.py", line 1, in <module>
    x = int(input())
ValueError: invalid literal for int()
with base 10: 'fjdsklfjsdf'
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```

```
$ python le.py < le.py
```

Programs and inputs

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```
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```



Nice programs

We can systematically modify any python program P so that

- ▶ P contains a main function that works with the input as a string.
- ▶ P never crashes. (If the original P crashes, the modified P outputs no.)

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```
x = int(input())
if x % 2 == 0:
    print('yes')
else:
    print('no')
```

```
import sys
def main(w):
    try:
        x = int(w)
        if x % 2 == 0:
            print('yes')
        else:
            print('no')
    except:
        print('no')
```

```
if __name__ == '__main__':
    w = sys.stdin.read()
    main(w)
```

When running a program

When you run a program P with input x , there are three possible outcomes:

- ▶ P terminates and outputs **yes**,
- ▶ P terminates and outputs **no**, and
- ▶ P does not terminate. (It runs forever.)

When running a program

When you run a program P with input x , there are three possible outcomes:

- ▶ P terminates and outputs **yes**,
- ▶ P terminates and outputs **no**, and
- ▶ P does not terminate. (It runs forever.)

Remarks: if P crashes (even after modification), we treat it as if it terminates and outputs **no**.

Proving impossibility

ជំនាញ L ត្រូវបានសម្រួល python P តែងតែដេក

P ដេក L

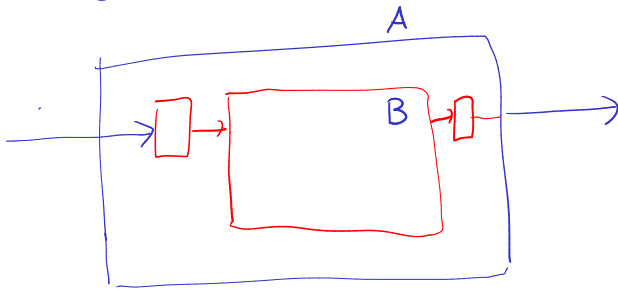
P decides L

- $\forall w \in L$, $P(w)$ output "yes"
- $\forall w \notin L$, $P(w)$ output "no"

→ មិនមាន loop → always terminate

Reduction: rough idea

— ifn subroutine.



▪ A decide P

↪ A for B ifn
- B decide Q

↪ B n decide Q

⇒ A n decide P

|||

↪ A n decide P ⇒

↪ B n decide Q

Language A

Let $\mathbb{P}$ be the set of all python programs. Let the language A be

$$\{P \in \mathbb{P} \mid \text{when running } P \text{ with } P \text{ as an input, } P \text{ terminates}\}$$

Problem: If I know P , can $P(P)$ terminate?

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We use a function call notation $P(x)$ when referring to the execution of program P with input x .

$$\{P \in \mathbb{P} \mid P(P) \text{ terminates}\}$$

ถามตัวเอง

ทำงาน

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We use a function call notation $P(x)$ when refering to the execution of program P with input x .

We restate the definition of A as

$$\{P \in \mathbb{P} \mid P(P) \text{ terminates}\}.$$

Deciders

We say that a python program P decides the language L if for any input string x , P when running with x as an input,

- ▶ P always terminates, 
- ▶ P outputs yes, if $x \in L$, and
- ▶ P outputs no, if $x \notin L$.

Deciders: more examples

Language A

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Is decider for A Is program P for
if P decides A

Not a decider for A

$P(P)$ term.

Input: python program P (as a string)

1. Load module P as $Pmod$
2. Call $Pmod.main(P)$
3. print('yes')

we reach this line,
only if $M.main(P)$ terminates

} in $P(P)$ loops
→ Unknown X

Lemma 2

There is no python program that decides A .

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We will see the proof at the end of class.

Undecidability

Computing / Computation



If we believe that anything that a computer can do can be written as a python program,

Undecidability

If we believe that anything that a computer can do can be written as a python program, and there is no python program that decides A , when we say that

A is undecidable.

Language A will be very important later on, we give it a proper name as SELFHALT.

$P(P) \dots \uparrow$

The proof as a table

List all python programs in $\mathbb{P}$ as $P_1, P_2, P_3, \dots$

	P_1	P_2	P_3	P_4	P_5	P_6	P_7	$\dots$
P_1								
P_2								
P_3								
P_4								
P_5								
P_6								
$\vdots$								
(B)								

What does B do on each input program P_i ?

Another language HALT

Let

$$\text{HALT} = \{(P, w) \mid P \text{ is a python program such that } P(w) \text{ terminates}\}$$

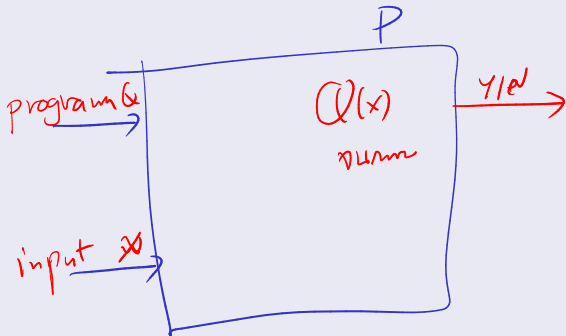
We shall prove that HALT is also undecidable (if we believe that python programs represent all possible computation).

Lemma 3

HALT is undecidable.

Proof.

We prove the lemma by contradiction. Assume that there is a python program P that decides HALT. *



Lemma 3

HALT is undecidable.

Proof.

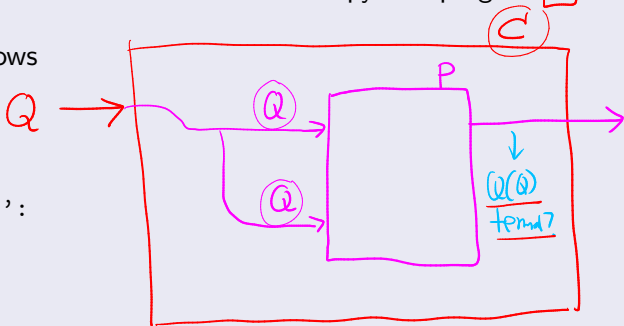
We prove the lemma by contradiction. Assume that there is a python program P that decides HALT.

We construct a program C as follows

Program C

Input Q

1. Load P as module $Pmod$
2. if $Pmod.main(Q,Q) == \text{'yes'}$:
3. print('yes')
4. else
5. print('no')



Proof.

We prove the lemma by contradiction. Assume that there is a python program P that decides HALT.

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Program C

Input Q

```
1.  if P(Q,Q) == 'yes':  
2.      print('yes')  
3.  else  
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```

Proof.

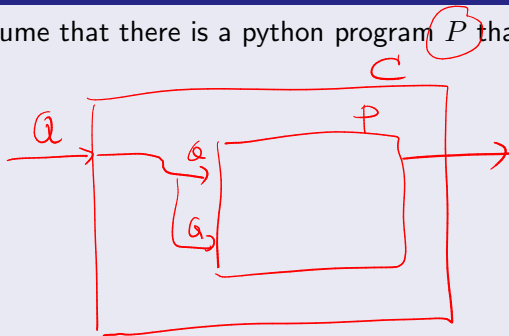
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Given program P , we can construct a program C that decides SELFHALT.

Proof.

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Program C

Input Q

```
1.  if P(Q,Q) == 'yes':  
2.      print('yes')  
3.  else  
4.      print('no')
```

Given program P , we can construct a program C that decides SELFHALT. However, we know that SELFHALT is undecidable; thus, we reach a contradiction.

We conclude that there does not exist a python program P that decides HALT. □

Reduction

ถ้าโปรแกรมที่ decide $A \Rightarrow$ โปรแกรมที่ decide B

① ถ้า A is decidable $\Rightarrow B$ is decidable.

ถ้า ② โปรแกรมที่ B is undecidable.

$\Rightarrow$ โปรแกรมที่

A is also undecidable

Reduction

reduction
|
ลดรูป

Goal: จ:พิสูจน์ว่า A undecidable :

- หาปัญหา B ที่ undecidable

- พิสูจน์ว่า ถ้า A decidable $\Rightarrow$ B decidable.

⋮

$\neg Q$
 $P \Rightarrow Q$
 $\neg P$

Reduction

- ▶ We show that if HALT is decidable, then SELFHALT is also decidable.

Reduction

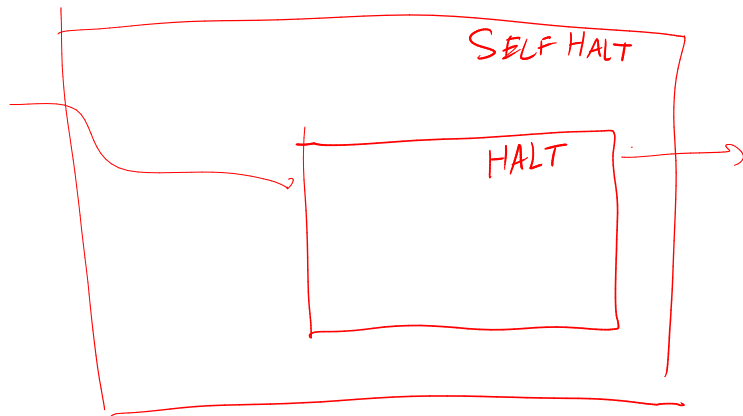
$$P \Rightarrow Q$$

- ▶ We show that if `HALT` is decidable, then `SELFHALT` is also decidable.
- ▶ However, `SELFHALT` IS UNDECIDABLE. $\neg Q$

Reduction

- ▶ We show that if HALT is decidable, then SELFHALT is also decidable.
- ▶ However, SELFHALT IS UNDECIDABLE.
- ▶ We conclude that HALT is also undecidable.

Reduction in picture



reduction on SELF HALT to HALT.

Let $\text{ACCEPT} = \{(P, w) \mid P \in \mathbb{P} \text{ and } P(w) \text{ terminates with acceptance}\}$.

Lemma 4

ACCEPT is undecidable.

Proof.

We prove the lemma by contradiction. Assume that there is a python program Q that decides ACCEPT.

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Program C

Input P, w

1. Replace every "print('no')" statement in P with "print('yes')"
1. if Q(P, w) == 'yes': // P(w) halts/terminate
2. print('yes')
3. else
4. print('no')

Yes: acc

No: loop
refw

Proof (cont.)

Program C $\rightarrow$ decides HALT.

Input P, w

1. Replace every "print('no')" statement in P^* with "print('yes')" $\Rightarrow P'$
1. if $Q(P', w) == \text{'yes'}$:
2. print('yes')
3. else
4. print('no')

We have to make sure that our reduction is correct by considering two cases.

Case 1: when $\underline{P(w)}$ halts.

$P'(w)$ prints yes $\Rightarrow Q(P', w) = \text{yes} \Rightarrow C$ prints yes ✓

Proof (cont.)

Program C

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We have to make sure that our reduction is correct by considering two cases.

Case 1: when $P(w)$ halts.

Case 2: when $P(w)$ does not halt.

$P(w) \text{ to accept } \Rightarrow Q(P, w) \neq \text{'yes'} \Rightarrow C \text{ no. } \checkmark$

Proof (cont.)

Program C

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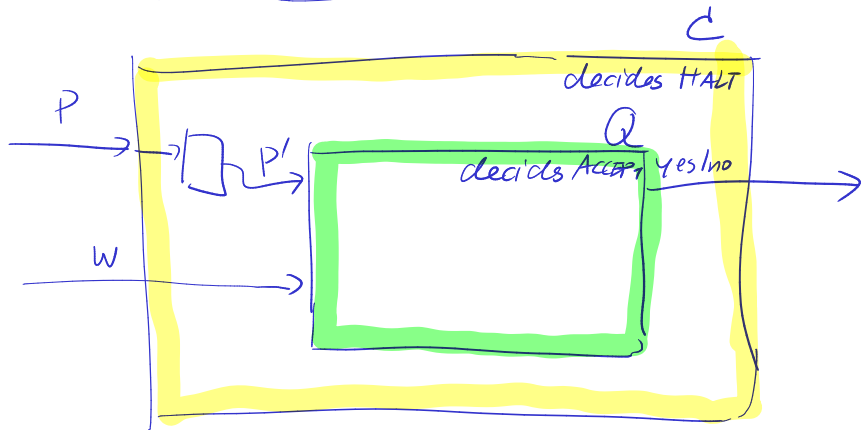
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Case 1: when $P(w)$ halts.

Case 2: when $P(w)$ does not halt.

Since in both cases, C answers correctly, we know that given program Q deciding ACCEPT, we can construct a program C that decides HALT. However, we know that HALT is undecidable; thus, we reach a contradiction. We conclude that ACCEPT is also undecidable. $\square$

Reduction from HALT to ACCEPT in picture



How about REJECT?

Let

$$\text{REJECT} = \{(P, w) \mid P \in \mathbb{P} \text{ and } P \text{ rejects } w\}.$$

Lemma 5

There is no python program that decides SELFHALT.

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We prove by contradiction. Assume that there is a python program P that decides SELFHALT.

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We describe a python program B that reads a string Q as an input as follows:

Program B

Input Q

```
1.    Load P as module Pmod
2.    if Pmod.main(Q) == 'yes':      # when Pmod outputs yes
3.        while True: pass           #    loop forever
4.    else:                           # when Pmod outputs no
5.        quit()                     #    halt
```

Given program Q as an input, B loops forever when

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Given program Q as an input, B loops forever when

It terminates when



Proof.

We know that

- ▶ $B(Q)$ loops when $Q(Q)$ terminates, and
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Since either $B(B)$ loops or terminates, and we cannot be in any of the cases, we obtain a contradiction.

Therefore, we conclude that program P does not exist.



Python as computation

Do you believe in this assumption:

Anything that a computer can do can be written as a python program.

Turing machines

Anything that a computer can do can be carried out using Turing machines.

Turing machines

Anything that a computer can do can be carried out using Turing machines.

Any possible computation can be performed by Turing machines.